

DPP NO.-9

- Q.1 $\sum_{r=0}^{n-1} \frac{{}^nC_r}{{}^nC_r + {}^nC_{r+1}}$ is equal to
(A) $\frac{n(n-1)}{2(n+1)}$ (B) $\frac{n+1}{2}$ (C) $\frac{n(n+1)}{2}$ (D) $\frac{n}{2}$
- Q.2 Let m denote the number of ways in which 4 different books are distributed among 10 persons, each receiving none or one only and let n denote the number of ways of distribution if the books are all alike. Then :
(A) $m = 4n$ (B) $n = 4m$ (C) $m = 24n$ (D) none
- Q.3 The number of ways in which we can arrange n ladies & n gentlemen at a round table so that 2 ladies or 2 gentlemen may not sit next to one another is :
(A) $(n-1)!(n-2)!$ (B) $(n!)(n-1)!$
(C) $(n+1)!(n)!$ (D) none
- Q.4 There are six periods in each working day of a school. Number of ways in which 5 subjects can be arranged if each subject is allotted at least one period and no period remains vacant is
(A) 210 (B) 1800 (C) 360 (D) 120
- Q.5 The number of all possible selections of one or more questions from 10 given questions, each question having an alternative is :
(A) 3^{10} (B) $2^{10} - 1$ (C) $3^{10} - 1$ (D) 2^{10}
- Q.6 A team of 8 students goes on an excursion, in two cars, of which one can seat 5 and the other only 4. If internal arrangement inside the car does not matter then the number of ways in which they can travel, is
(A) 91 (B) 182 (C) 126 (D) 3920
- Q.7 The number of divisors of the number 21600 is _____ and the sum of these divisors is _____.
- Q.8 10 IIT & 2 PET students sit in a row. The number of ways in which exactly 3 IIT students sit between 2 PET student is _____.
- Q.9 The number of ways of choosing a committee of 2 women & 3 men from 5 women & 6 men, if Mr. A refuses to serve on the committee if Mr. B is a member & Mr. B can only serve, if Miss C is the member of the committee, is :
(A) 60 (B) 84 (C) 124 (D) none
- Q.10 Six persons A, B, C, D, E and F are to be seated at a circular table. The number of ways this can be done if A must have either B or C on his right and B must have either C or D on his right is :
(A) 36 (B) 12 (C) 24 (D) 18

- Q.11 There are 2 identical white balls, 3 identical red balls and 4 green balls of different shades. The number of ways in which they can be arranged in a row so that atleast one ball is separated from the balls of the same colour, is :
 (A) $6(7! - 4!)$ (B) $7(6! - 4!)$ (C) $8! - 5!$ (D) none
- Q.12 Sameer has to make a telephone call to his friend Harish, Unfortunately he does not remember the 7 digit phone number. But he remembers that the first three digits are 635 or 674, the number is odd and there is exactly one 9 in the number. The maximum number of trials that Sameer has to make to be successful is
 (A) 10,000 (B) 3402 (C) 3200 (D) 5000
- Q.13 Six people are going to sit in a row on a bench. A and B are adjacent, C does not want to sit adjacent to D. E and F can sit anywhere. Number of ways in which these six people can be seated, is
 (A) 200 (B) 144 (C) 120 (D) 56
- Q.14 Given 11 points, of which 5 lie on one circle, other than these 5, no 4 lie on one circle . Then the maximum number of circles that can be drawn so that each contains atleast three of the given points is :
 (A) 216 (B) 156 (C) 172 (D) none
- Q.15 One hundred management students who read at least one of the three business magazines are surveyed to study the readership pattern. It is found that 80 read Business India, 50 read Business world, and 30 read Business Today. Five students read all the three magazines. How many read exactly two magazines?
 (A) 50 (B) 10 (C) 95 (D) 25
- Q.16 Find the number of 10 digit numbers using the digits 0, 1, 2, 9 without repetition. How many of these are divisible by 4.
- Q.17 A four digit number is called a doublet if any of its digit is the same as only one neighbour. For example, 1221 is a doublet but 1222 is not. Number of such doublets are
 (A) 2259 (B) 2268 (C) 2277 (D) 2349